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Lab Report

Design of Second-Order Continuous Time Sigma-Delta Modulator in 40nm CMOS

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Contents

1 Delta Sigma ADC design

1.1 Spec

Table 1.1: Performance Comparison

Parameter	**Target Spec**	**Achieved Spec (nom)**
Input voltage range	$\pm$300 mV	$\pm$300 mV
Signal Bandwidth	100 kHz	100 kHz
Supply voltage (V_{dd})	0.55 V	$0.55 \pm 10\%$ V
Ground voltage (VSS)	-0.55 V	$-0.55 \pm 10\%$ V
OSR	–	256
Required SNR (dB)	65	63.57
Power	As low as possible	230 μW
ENOB (Bits)	10.5	10.27

1.2 Tasks

1.2.1 NTF for 1st and 2nd Order Modulators MATLAB

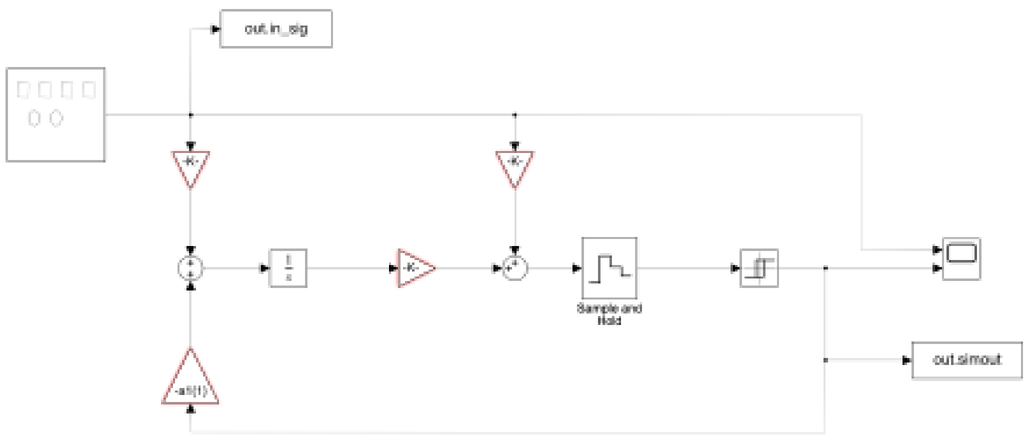

Figure 1.1: First order schematic

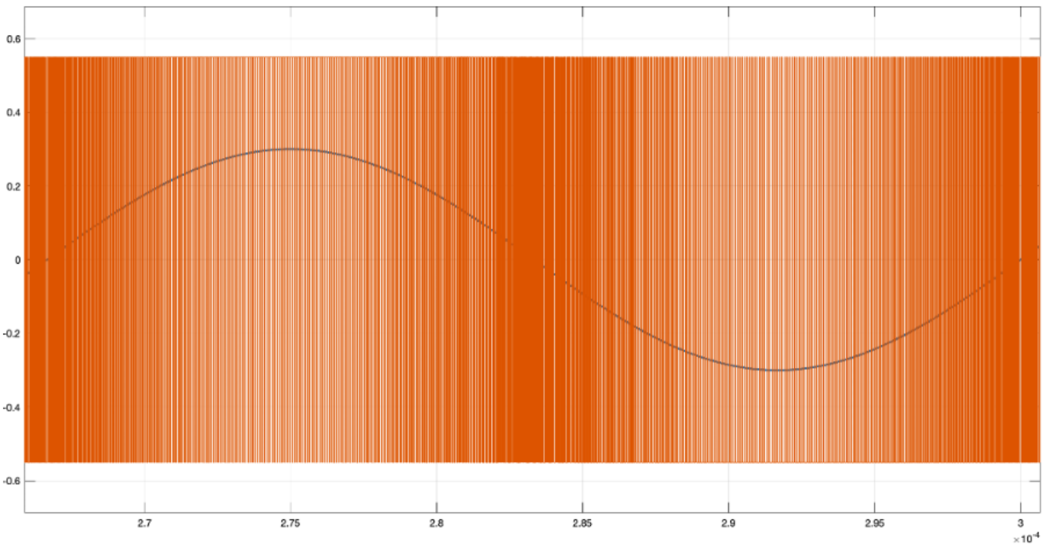

Figure 1.2: First order output wave

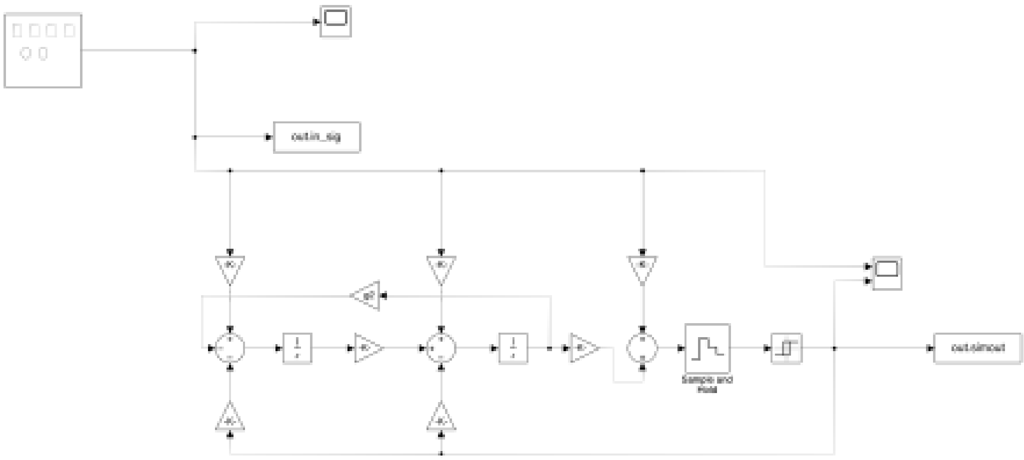

Figure 1.3: Second order schematic

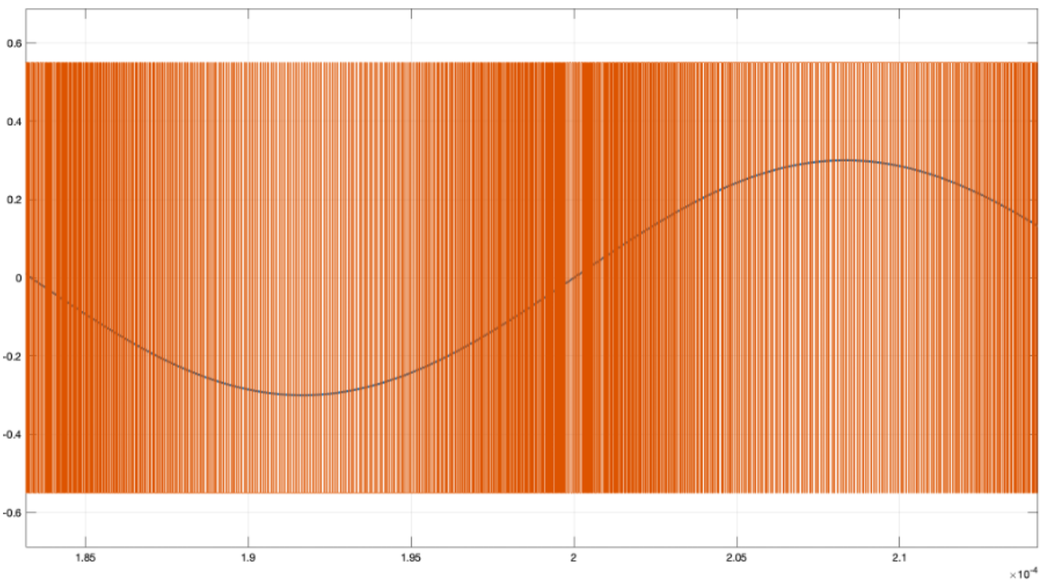

Figure 1.4: Second order output wave

First order: $\frac{Z-1}{Z-0.333}$ Second order: $\frac{z^2-2z+1}{z^2-1.225z+0.4415}$ In Matlab code I simulate both 1st order and 2nd order noise shaping 1.5 1.7, with out of band gain 1.5. Although we will low pass our noise with decimation filter at the end, it's still good to set out of band gain limit to ensure that the high frequency components doesn't affect maximum stable amplitude (MSA).

We can see from the schematic1.1 1.3 that this is a CIFB structure. K is the parameter calculated using DStoolbox, and $\frac{1}{S}$ represents integration. The equation for the integrator is

$$\frac{1}{1-Z^{-1}}$$

Approximating Z as $1+ST$, the integration formula simplifies to

$$\frac{1}{ST}$$

Since $T=\frac{1}{f_s}$, we need to multiply by f_s after performing the $\frac{1}{s}$ integration.

1.2.2 FFT in the Frequency Domain (0 to fs/2), Describe and Interpret Your Result

From FFT plot 1.6 1.8 it's clear that 2nd order has better noise shaping, it's steeper at high frequencies (approx.. 40db/dec) whereas 1st order modulator only has approx.. 20db/dec. Also, second order modulator push in-band noise to lower value.

We can also look at ideal implementation output waveform 1.2 1.4, we can observe more transitions at lower input voltages, which means ADC is switching more between 1 and 0. At high and low peak, the output will produce series of 1's and 0's.

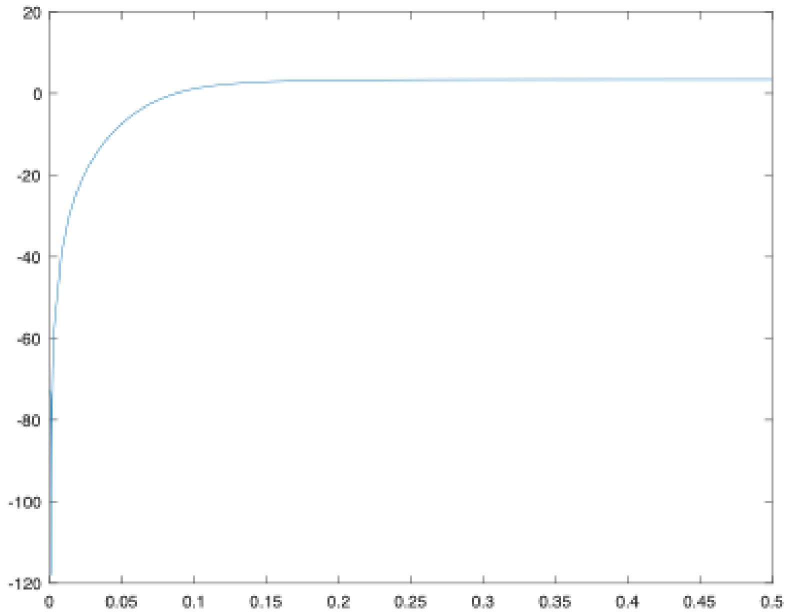

Figure 1.5: First order NTF

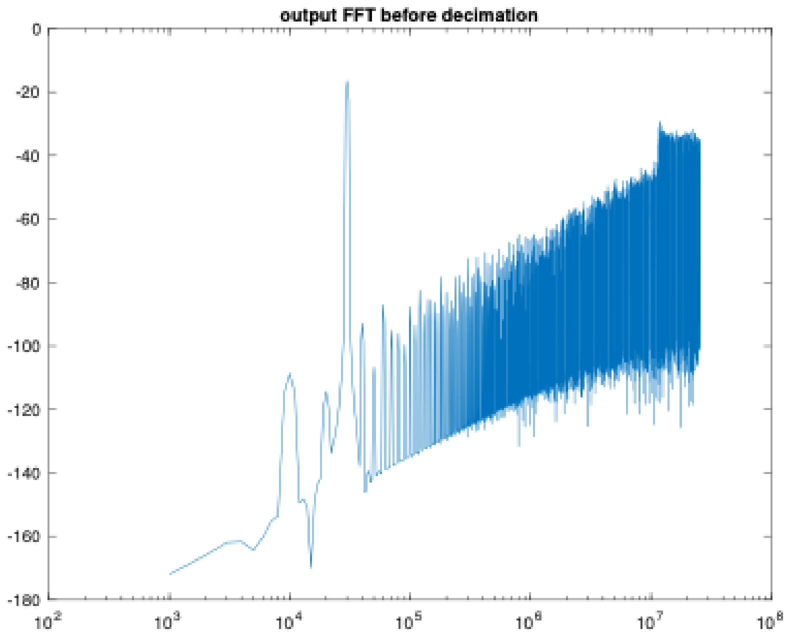

Figure 1.6: First order FFT

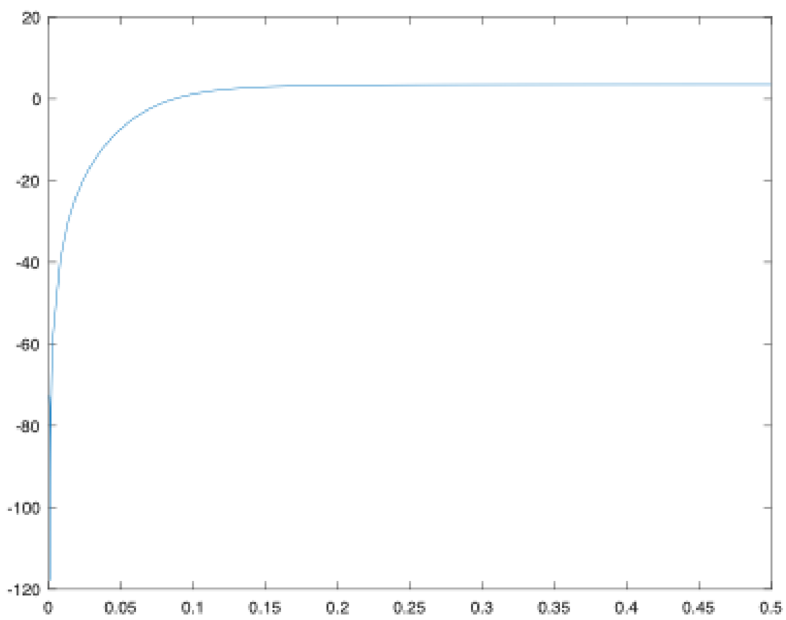

Figure 1.7: Second order NTF

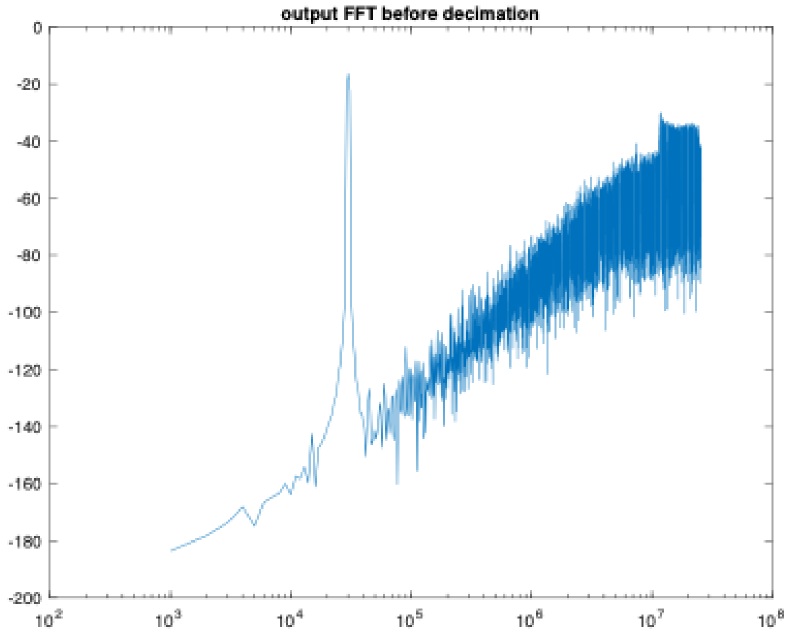

Figure 1.8: Second order FFT

1.2.3 Use a constant input signal Vin = 1/3 Vref. Simulate the system and analyze the output signal in the time and frequency domain. What undesired effects occur and why do they occur? Are they similar in both 1st and 2nd order modulator? Why or why not?

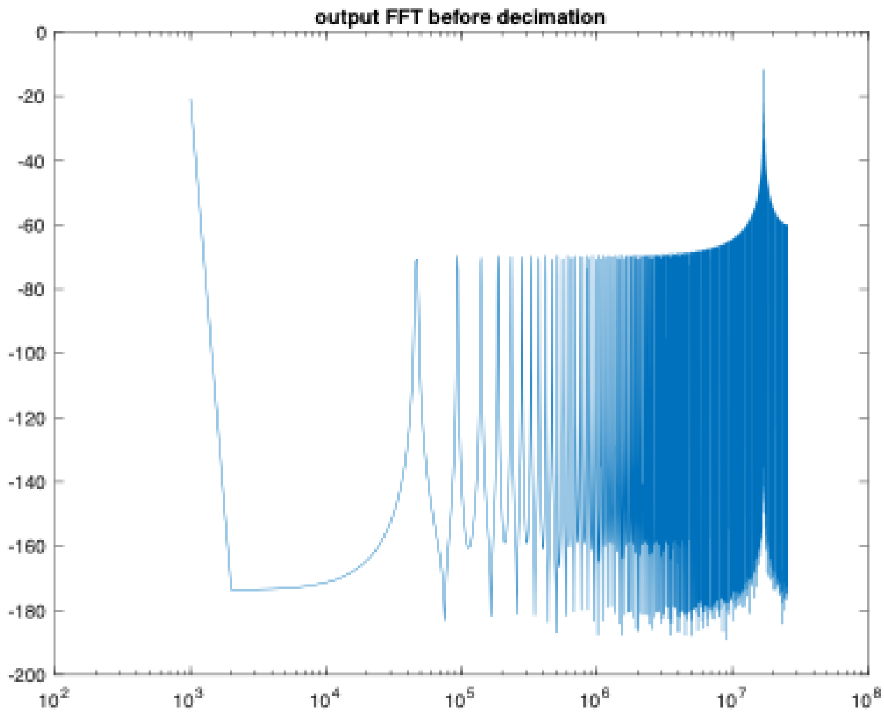

Figure 1.9: First order FFT result, input = 1/3 Vref

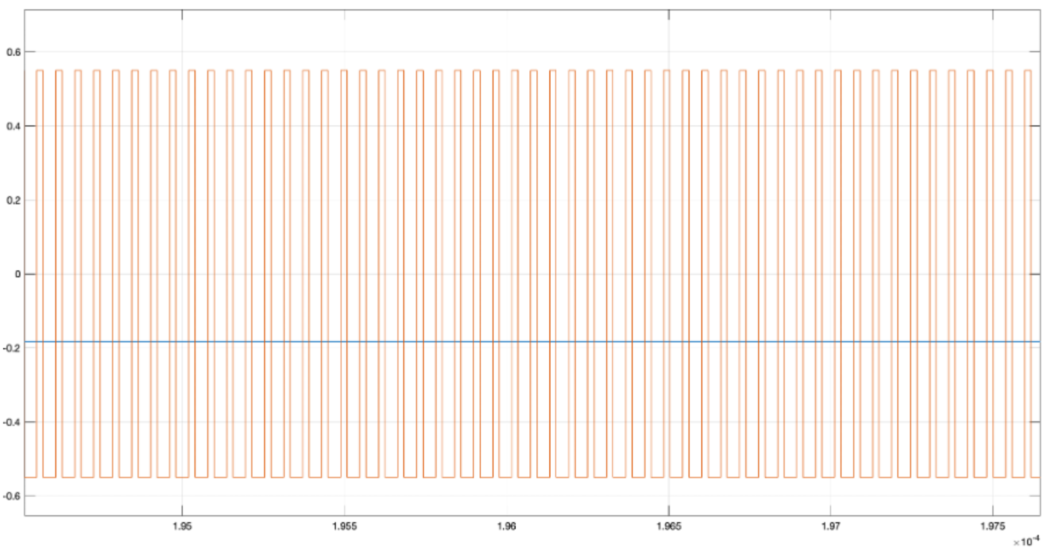

Figure 1.10: First order output wave, input = 1/3 Vref

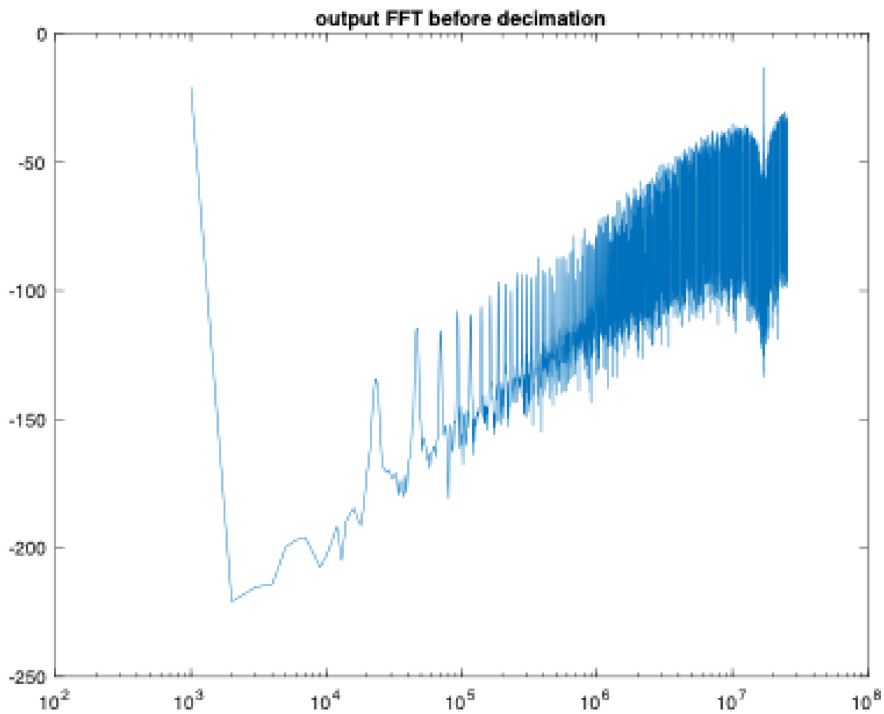

Figure 1.11: Second order FFT when vin = 1/3 vref

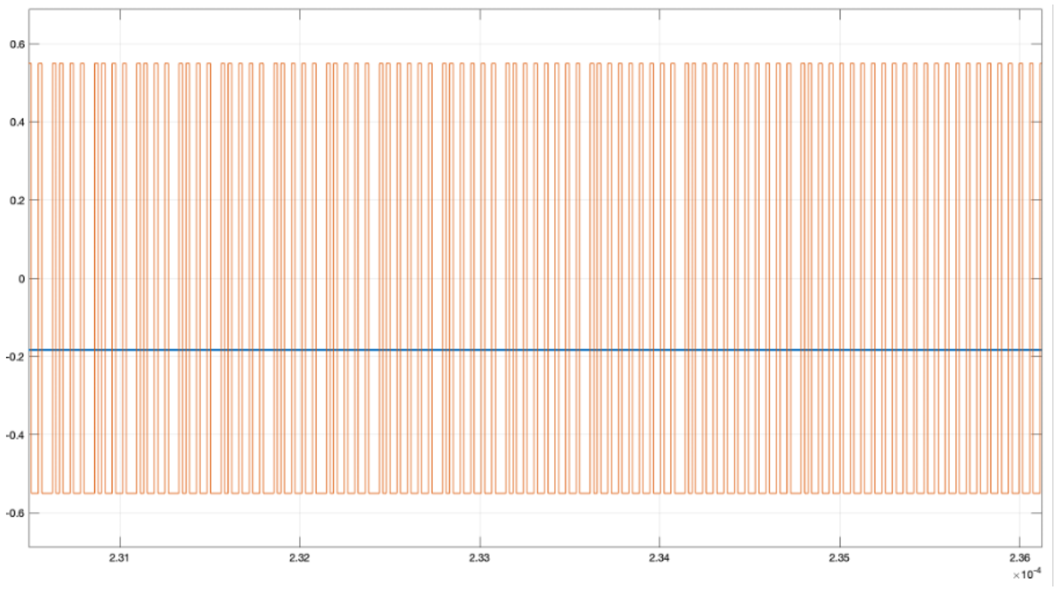

Figure 1.12: Second order output wave when vin = 1/3 vref

In first order system1.10, we can see exactly 1/3 of output is 1. In second order system1.12 we can see the patterns are not same as first order. This is because in first order converter, there is only 1 integral path. Hence the duty ratio will be 1/3, which is equal to the input we feed. First order is ideally a pulse density modulator. Second order converter, there are 2 integral paths, hence we will not obtain constant duty ratio.

If we see FFT waveform. In first order 1.9 we can see a peak at approx. our sampling frequency. In second order FFT 1.12, the noise shaping is more obvious, we have also a peak at sampling frequency. In both graphs we can see the low frequency component produced by our input dc signal.

1.2.4 Implement the design of a Continuous time second order Delta-Sigma modulator in Cadence Virtuoso with real components

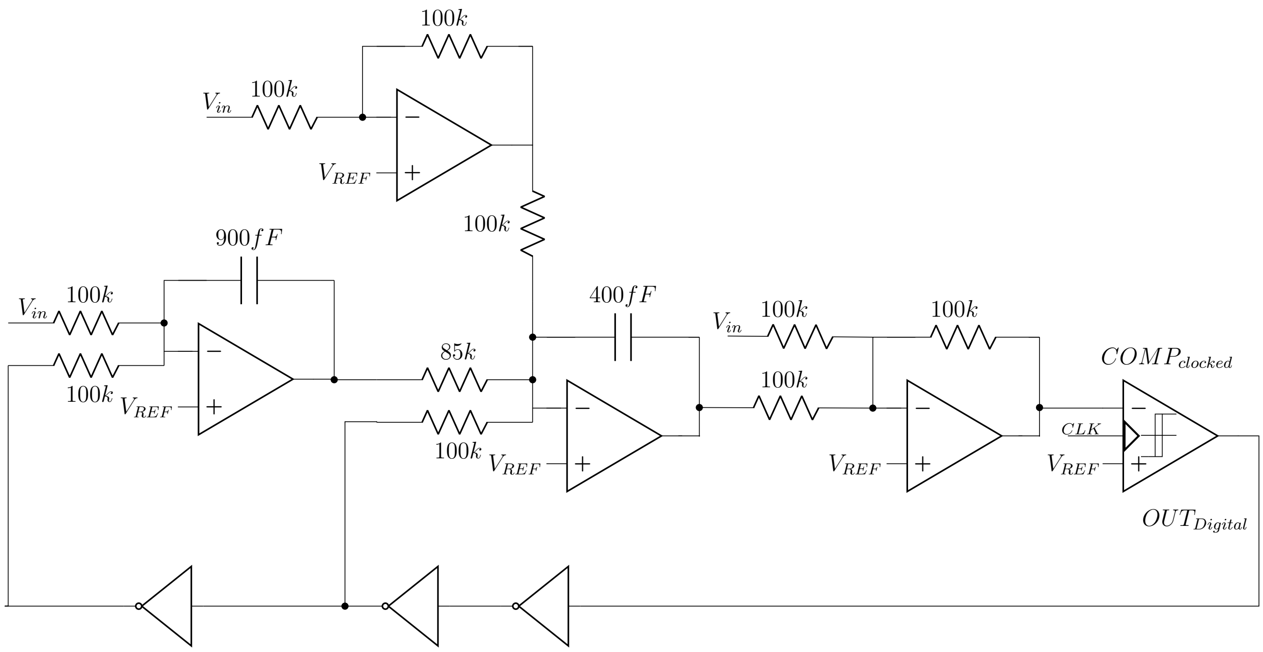

Figure 1.13: Top Level Schematic of the Delta-Sigma Modulator

1.2.4.1 Top Level Implementation of the Sigma-Delta Modulator in Cadence Virtuoso

The top level schematic of the second order implementation is shown in Fig.1.13. The architecture has two integrators and two summing amplifiers to implement the CIFB loop, totalling a total of four opamps in the design. This has given us considerable energy savings in terms of power. The system also includes a clocked comparator which models the 1-bit quantizer. In the feedback loop, we have added inverters that have the capability to drive the resistors used in the feedback path of the integrator and summer.

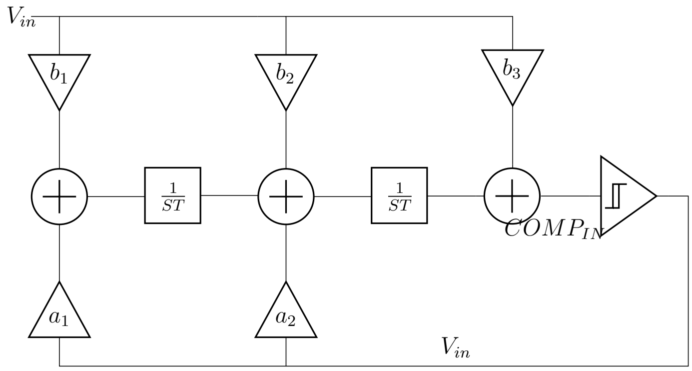

Figure 1.14: Structure of the CIFB architecture

The critical design parameter at the top level of the converter is choice of resistors and the capacitors. The second order CIFB structure is shown in Fig.1.14. The $COMP_{IN}$ can be written as,

$$COMP_{IN} = V_{in}[b_3 + \frac{b_2}{sT} + \frac{b_1}{(ST)^2}] - V_{fb}[\frac{a_2}{ST} + \frac{a_1}{(ST)^2}] \tag{1.1}$$

Let's consider the first summing node, the gain of V_{in} in this node is given by,

$$\frac{V_{out}}{V_{in}} = \frac{b_1}{ST} \tag{1.2}$$

Where, $\frac{1}{2*OSR*F_{BW}}$.
The gain of the individual path in the loop can be modeled as the gain of the integrator which is given by,

$$\frac{V_{out}}{V_{in}} = \frac{1}{SRC} \tag{1.3}$$

From Eq.1.2 and Eq.1.3, we get,

$$T = \frac{RC}{b_1} \tag{1.4}$$

The value of R was chosen as $100K$ to minimize thermal noise and the value of C was obtained as, 1 Similarly all the remaining values of resistors and capacitors were estimated. The waveform obtained from the real implementation of the converter is shown in Fig.1.15. Clearly, the output waveform shows identical characteristics to the ideal implementation in MATLAB.

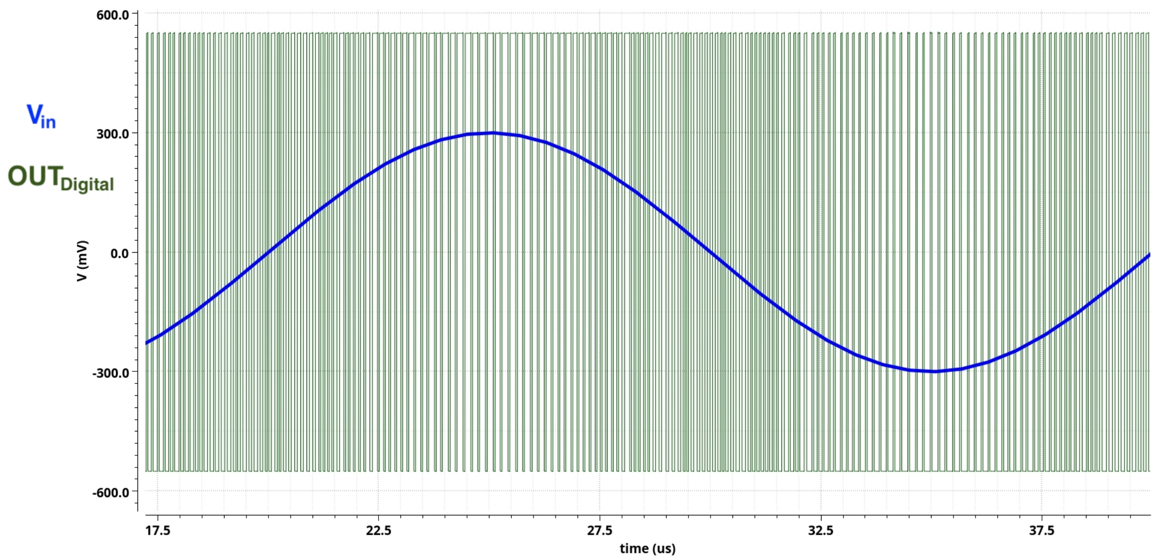

Figure 1.15: Output Waveform of the 2^{nd} order Δ-Σ Modulator Designed in Virtuoso

To calculate the spectral characteristics of the converter, we use the Spectrum Analyzer from Virtuoso. The input frequency of the signal is 50KHz and total run time for the simulation 320us to achieve coherent sampling and thereby avoid spectral leakage while calculating the FFT of the signal. The typical output spectrum is shown in Fig.1.16. The key design metrics of the converter is depicted in Table 1.2

Parameter	**SNR (dB)**	**ENOB**	**SFDR (dB)**	**Power (µW)**
Nominal	63.57	10.27	64.14	230

Table 1.2: Performance Metrics

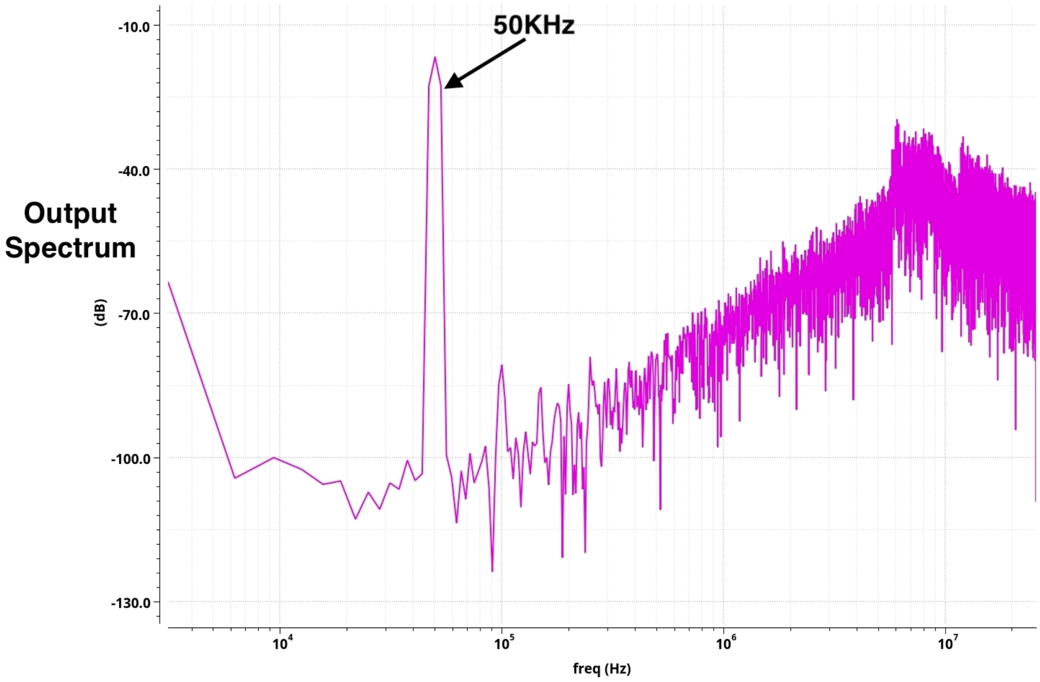

Figure 1.16: FFT Spectrum of the 2^{nd} order Δ-Σ Modulator Designed in Virtuoso

1.2.4.2 Implementation and Design of Opamp for Integrator

The integrator accumulates the input signal over time, effectively shaping the quantization noise.It pushes the quantization noise to higher frequencies, where it can be removed by a digital low-pass filter.
In order to design a proper Opamp as a part of the integrator, a rail to rail topology is proposed. Considering the low battery voltage (1.1V), monte-celli stage is replaced by self biased topology as a trade of for current control as Fig. 1.17 shown.

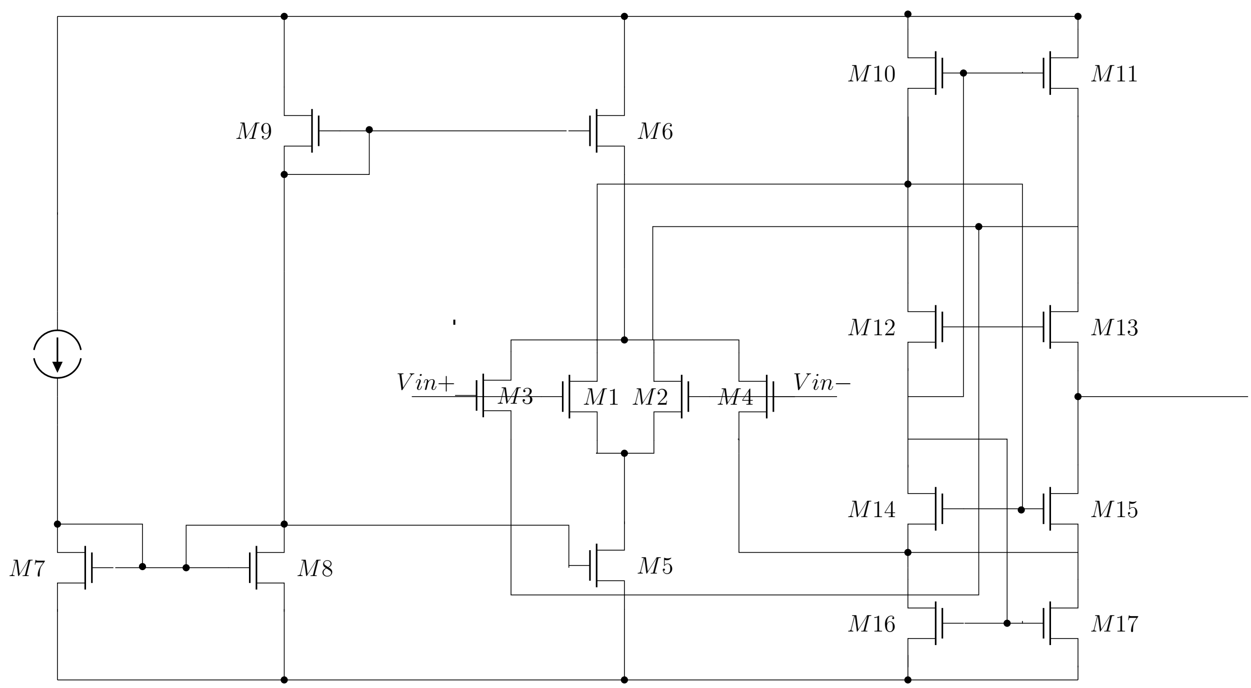

Figure 1.17: Schematic of the Self-biased Cascode

M1 ∼ M4 are the complementary MOSFETS pair to cover the rail to rail input range. M10 ∼ M17 are the load stage as folded cascode current mirror. Considering the low battery voltage, it is quite chalenging for biasing.

This topology has a benefit that can adjust rail to rail common mode voltage while keeping all transistors into saturation. Furthermore it doesn't require any bias circuit for the load stage. In the first topolgy an inverter is used as output stage for larger gain. This output inverter are sized large for strong current driving ability. In order to have a large unit gain bandwidth, the output stage is abandoned in order to have a rapid response. In the end, a folded cascode single stage is designed for rail to rail application.

The AC simulation result shown in Fig.1.18 indicates that this opamp have a DC gain of 40dB when common mode is 0V, and a 50 degree phase margin with UGB 518Mhz.

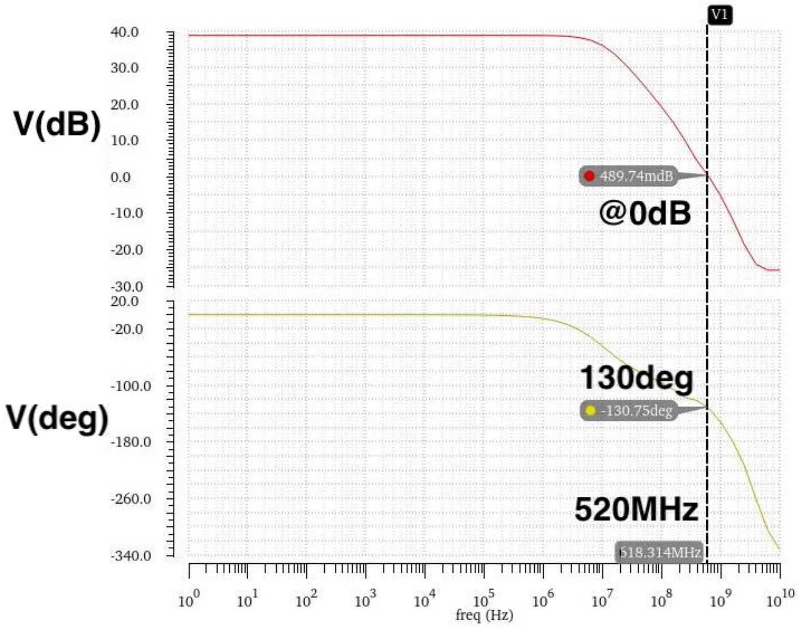

Figure 1.18: Gain and Phase Plot of the Opamp

As a result, the integrator in the ADC works as expected.

1.2.4.3 Comparator

Circuit Structure

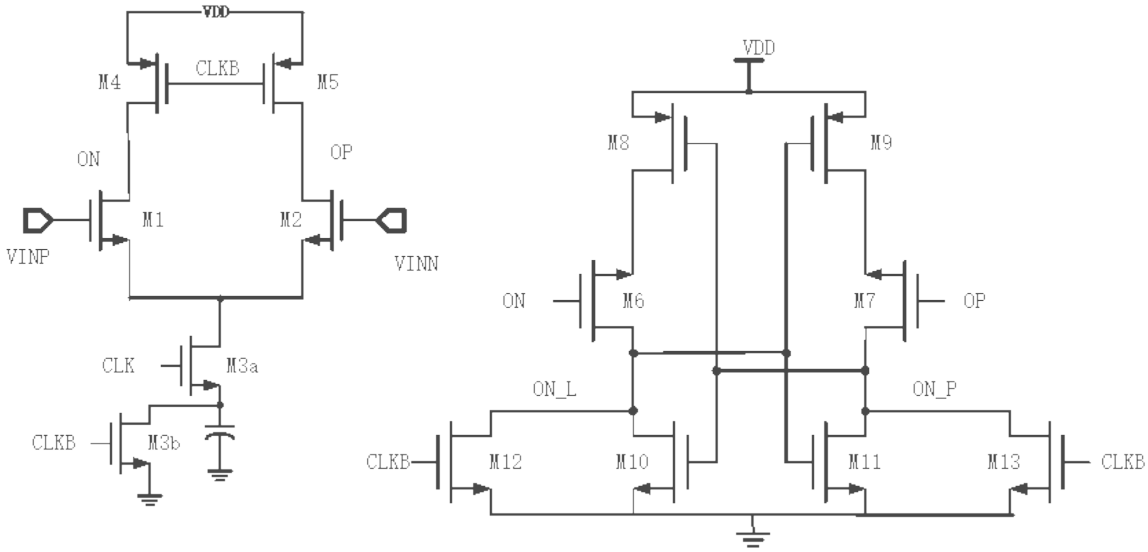

Figure 1.19: Comparator circuit structure

Operation Mode of Dynamic Bias Comparator

1. Reset Phase

- **Precharge**: Transistors M4/M5 charge the preamplifier output node Di+/Di- to V_{DD}.
- **Latch reset**: M12/M13 reset the latch output (Out+/Out-) to a low level.
- **Tail capacitor discharge**: The tail capacitor C_{tail} is discharged to GND through M3b.

2. Comparison Phase

- **Dynamic bias start**: M3a is turned on, Di+/Di- nodes charge C_{tail} through M1/M2, and the V_{cap} voltage gradually rises.
- **Adaptive working area**:
  - *Strong inversion region*: Initial high tail current achieves fast response (large signal input).
  - *Weak inversion region*: Tail current decreases, gain increases, and noise bandwidth decreases (small signal input).
- **Termination condition**: When V_{cap} rises to turn off one of M1/M2, the Di+/Di- node voltage freezes and the latch completes the regeneration output.

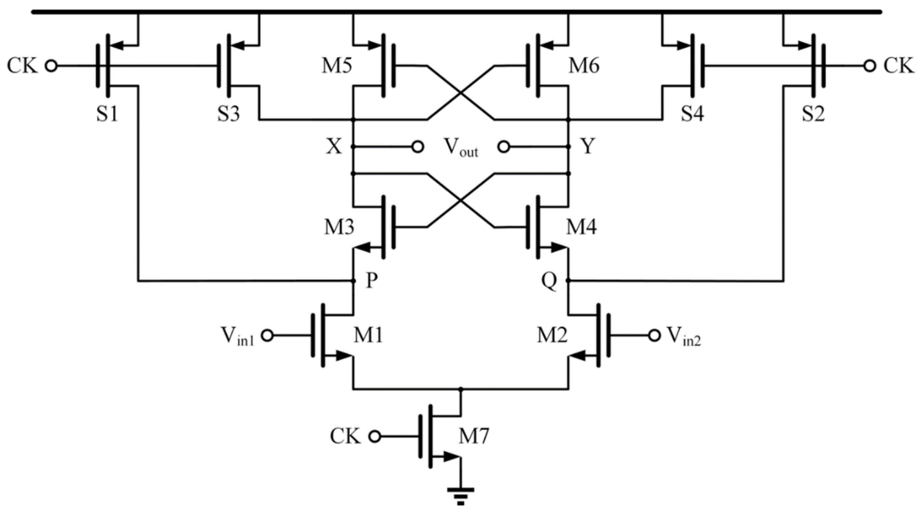

Figure 1.20: Design consistency with group implementations

Offset Characterization

Test methodology:

1. Input a ramp signal with **100 μV resolution**
2. Convert to differential signal using an ideal balun
3. Observe output with an ideal JK latch

Simulation results (non-Monte Carlo):

Measured offset = 0 μV

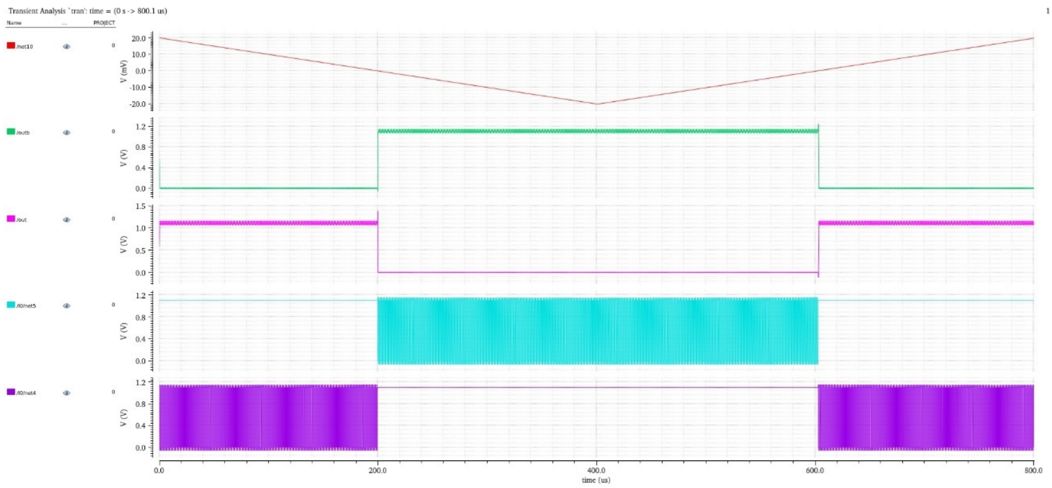

Figure 1.21: Offset distribution from Monte Carlo simulation

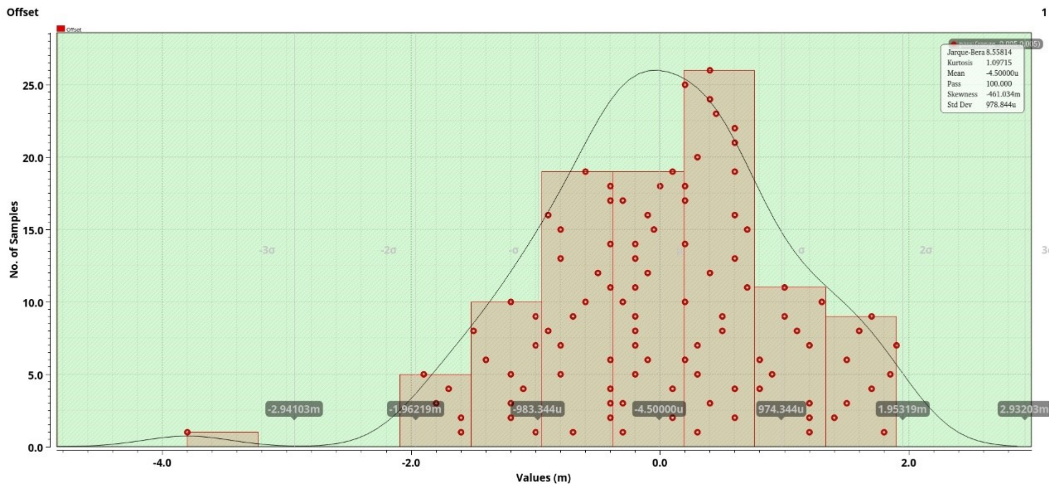

Figure 1.22: Offset distribution from Monte Carlo simulation

Noise Analysis
PSS/Pnoise simulation results:

$$\text{Input-referred noise} = \boxed{576.3\ \mu\text{V}_{\text{rms}}}$$

$$\text{Integrated noise} = \sqrt{\int_{f_{\min}}^{f_{\max}} S_{\text{noise}}(f)\, df}$$

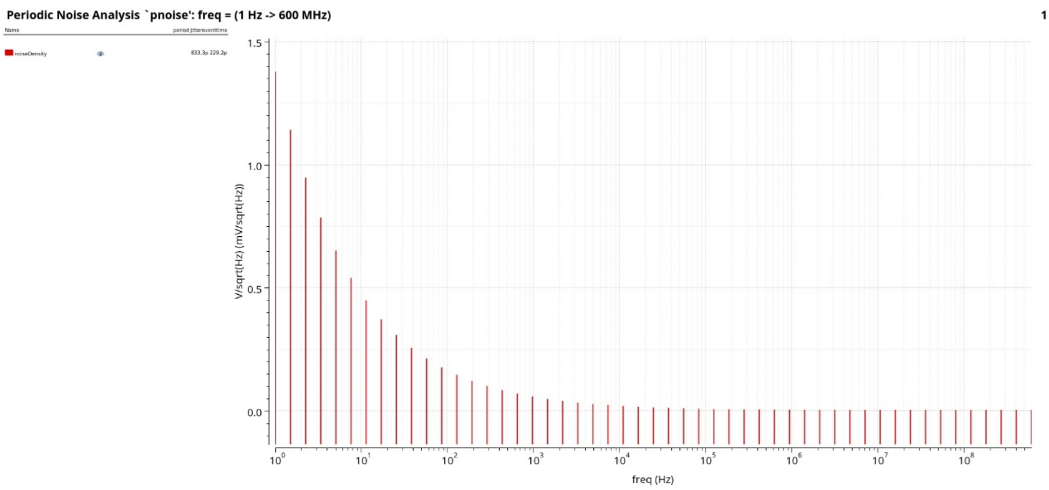

Figure 1.23: Input-referred noise spectral density

1.2.4.4 Digital Decimation filter

Introduction
Sigma-Delta ADC use oversampling and noise shaping to achieve high-resolution digital outputs. The raw output of a Sigma-Delta modulator is a high-frequency 1-bit bitstream, which must be processed through a decimation filter to extract lower-rate digital values. Therefore, the output data rate needs to be downsampled from the oversampling rate OSR×fs to the Nyquist frequency fs. In this process, redundant data generated by oversampling is eliminated and the resolution is increased, while ensuring the accuracy of data. **Design Concept**
The fundamental idea behind this decimation filter is to use counters to accumulate and average the bitstream output over a given time window. We use two 10-bit counters that have the following functions Counter 1: Counts a fixed number of clock

cycles to define the averaging period. Counter 2: Counts the number of logics ’1’s in the Sigma-Delta modulator output during these 1024 clock cycles. The output of the Modulator may contain multiple consecutive cycles of high-level signals. In order to achieve data separation, the output signals are converted from consecutive logic ‘1’s in multiple cycles to independent logic 1 outputs in each clock cycle by using an AND gate with the clock signals to ensure the signals are correctly synchronized and parsed, the output is used as input to counter 2 to calculate the number of logics ‘1’s in a counting cycle. The output of the Counter2 is quantized by the Ideal 10-bit ADC and compared to the input signal from the Sigma-Delta ADC to assess the accuracy and consistency of the signal.

Implementation Details

1 Clock Cycle Counter (Counter 1) A 10-bit counter increments every clock cycle. It is used to count 1024 CLK cycles to determine a complete sampling period. When the count reaches 1024, it will generate a reset signal to reset the counter 2 to complete a counting cycle. 2 Logics ‘1’Counter(Counter 2) A second counter increments only when the modulator output is logic ’1’.This counter accumulates the number of logics ’1’s observed over 1024 cycles. 3 Reset mechanism In the initial design, the reset was triggered when all 10 outputs of the first counter were ’1’s at the end of a counting cycle, I use ten AND gates to achieve it. However, this method introduced significant glitches. Based on calculations, if OSR is 128, the total counting cycle time of 1024 clock cycles is expected to be 40µs, but glitches occurring at 30µs interfere with the count, leading to inaccuracies. To mitigate this issue, an alternative reset mechanism is required to ensure precise counting and reliable data processing. So instead, the design was modified to detect the falling edge of the last output of the counter 1 to determine when 1024 CLK cycles have been completed. For counter 1, a 10-bit counter increments with each clock cycle and automatically resets upon reaching 1024, initiating a new counting sequence. And the reset signal for Counter 2 is controlled via an OR gate, which consists of two signals: Initialization Reset: Resets the counter when the system is initialized. Cycle Completion Reset: Resets the counter at the end of a complete counting cycle, preparing it for the next cycle. In order to prevent the reset signal from working too fast and causing the output result to always be 0, the reset signal should be delayed by a 4-stage inverter to ensure that the reset signal arrives at counter 2 one step later, so as to ensure that the counter works normally. 4 D- Flipflop The ideal flip-flops available in the AHDL library do not support reset functionality. Implementing a 10-

bit counter using these flip-flops would require additional circuitry to reset each input bit simultaneously, leading to increased circuit complexity and potential timing delays that affect output accuracy. To address this, a D-flip-flop with reset functionality was designed using NAND gates, ensuring a simpler and more reliable reset process while maintaining precise timing control. 5 10-bit Counter A 10-bit counter can be constructed using D flip-flops, where the logical expressions for each bit can be derived from a truth table to build a complete circuit. The counter output exhibits a clear binary increment pattern, with each bit's transition determined by the state of the preceding bit. The output of the previous flip-flop and the output of the current flip-flop are processed by the NAND Gate and then used as the input signal of the current flip-flop. 6 Output The output of the counter is stabilized using a 10-bit D-Flip-Flop, ensuring that the result remains steady throughout a full counting cycle. This prevents reset-induced amplitude recounts from 0. As a result, the waveform output remains consistent and accurately reflects the input signal characteristics. The final output is then processed through a 10-bit ideal ADC to obtain the output waveform.

Problem

The first sampling point, as the Modulator produces a continuous period of logics 1 outputs in the initial phase, may be affected in some way, resulting in a deviated waveform. However, in subsequent cycles, the signal gradually remains stable and subsequent results are not affected by this initial state and the decimation filter will operate normally. This the simplest decimation filter is more suitable for low frequency input signals. According to the previous calculation, if the OSR is 256 and the bandwidth is 100kHz, each counting cycle (i.e., the time to sample 1 point) is about 20µs. When the input signal frequency is 50kHz, the cycle time is also 20µs, which means that only 1 data point can be captured in a complete counting cycle, and the original waveform cannot be effectively reconstructed. When the input signal frequency is lower, such as 1kHz, and its period is 1ms, under the same decimation setting, 50 points can be sampled in each cycle, which can more accurately restore the input waveform and make the filtered signal closer to the original signal shape.

Optimization results

We can use Rolling Average instead of Floating Average. The Floating Average method results in high-frequency signals being sampled at only one point per cycle, leading to information loss. However, Rolling Average is a method that smooths data continuously. It calculates the average of the previous N samples at each new sampling point and then shifts the sliding window forward by one point to continue the computation. This way, even if the input signal frequency is high, multiple points are still retained per cycle, enhancing the signal reconstruction capability. In the process of rolling average computation, a new issue arises: if the earliest data in the sliding window is not discarded, the cumulative calculation result will continuously increase, affecting the accuracy and stability of the signal. To avoid this problem, the correct implementation is to subtract the oldest data while introducing new data, ensuring that the sliding window always processes a fixed number of sampling points. This approach not only effectively smooths the signal but also enhances signal reconstruction capability. In practical applications, we can introduce the CIC (Cascaded Integrator-Comb) filter to efficiently perform this computation.

CIC Decimation Filter

A CIC (Cascaded Integrator-Comb) filter is an FIR digital filter. Its key advantage lies in its simple structure: compared to conventional FIR filters, it eliminates the need for numerous multipliers by utilizing only adders and delays, significantly simplifying computations. Additionally, it requires no memory to store filter coefficients. However, its drawback is the inability to flexibly design amplitude-frequency characteristics, which sometimes necessitates adding a subsequent FIR filter for spectral shaping. The CIC filter comprises integrators and comb stages, and its cascade configuration enables its use as either a decimation or interpolation filter. The figure below illustrates the CIC decimation filter implemented in this experiment.

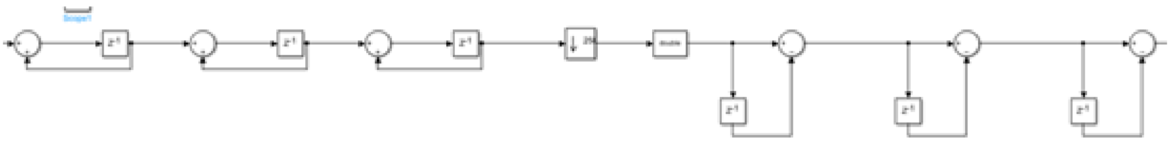

Figure 1.24: CIC filter structure inmatlab

The primary objective of the CIC filter is to convert the high-frequency single-bit code stream output from a Σ-Δ modulator into lower-frequency multi-bit quantized values.

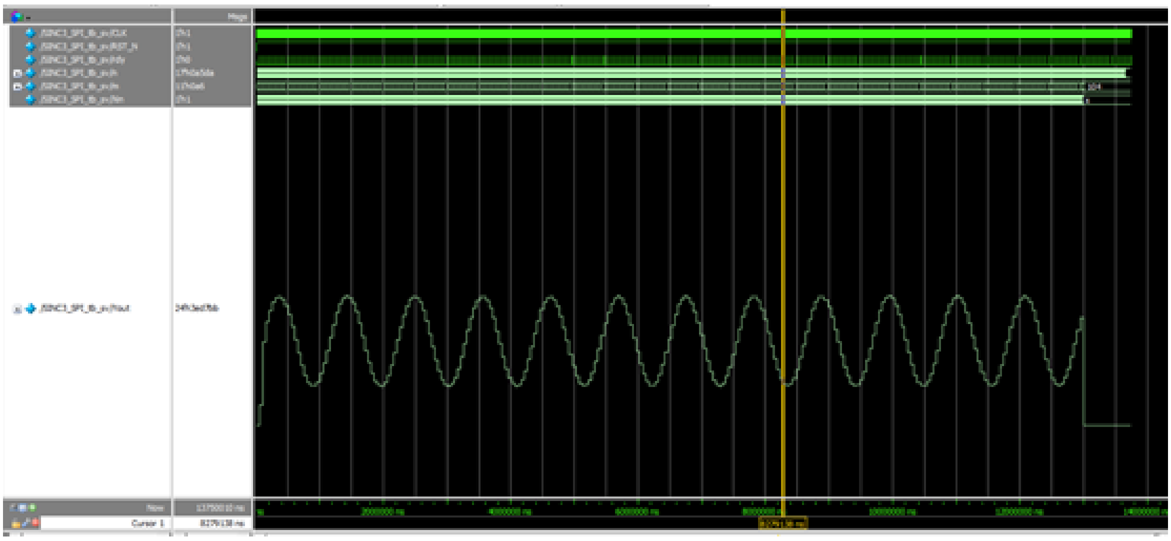

Figure 1.25: Filter simulation

In the figure above, signal X_{in} represents the high-frequency single-bit code stream from the Σ-Δ modulator, while Y_{out} denotes the multi-bit quantized output after CIC filtering.

Structure The CIC decimation filter fundamentally consists of integrators, a down-sampling decimator, and comb stages. Its transfer function is given by:

$$\mathrm{H}(z) = \left(\frac{1}{\mathrm{M}} \cdot \frac{1 - z^{-M}}{1 - z^{-1}} \right)^{K} \tag{1.5}$$

Frequency Response The amplitude-frequency response curve of a single-stage CIC filter is shown below, demonstrating its passband in the low-frequency region, thereby functioning as a low-pass filter.

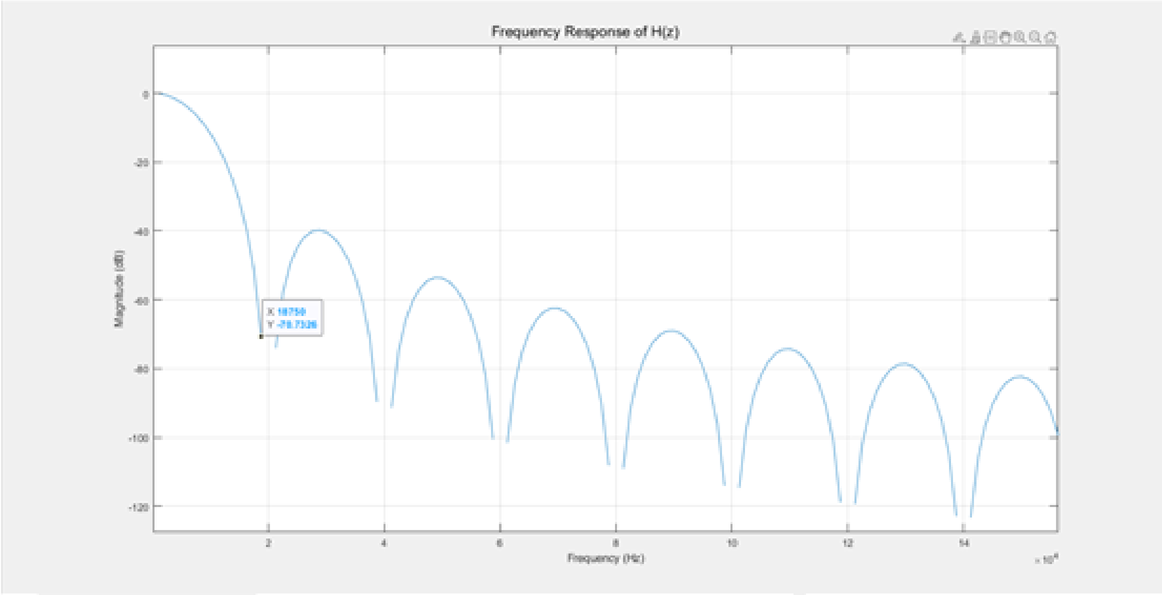

Figure 1.26: Frequency Response

The high-frequency single-bit stream from the Σ-Δ modulator is converted to multi-bit quantized values through this low-pass filtering. If the modulator's output is interpreted as a PWM waveform where the duty cycle represents the proportion of logic "1" in the conversion, the low-pass filter (with cutoff frequency below the fundamental harmonic of the square wave) preserves only the DC component. PWM waves with higher duty cycles retain greater energy and are converted to higher-amplitude DC values, while those with lower duty cycles yield lower-amplitude DC values, achieving single-bit to multi-bit conversion.

Results The filtered binary values produce a waveform shown below, which in MATLAB can be restored to a sine wave consistent with the input:

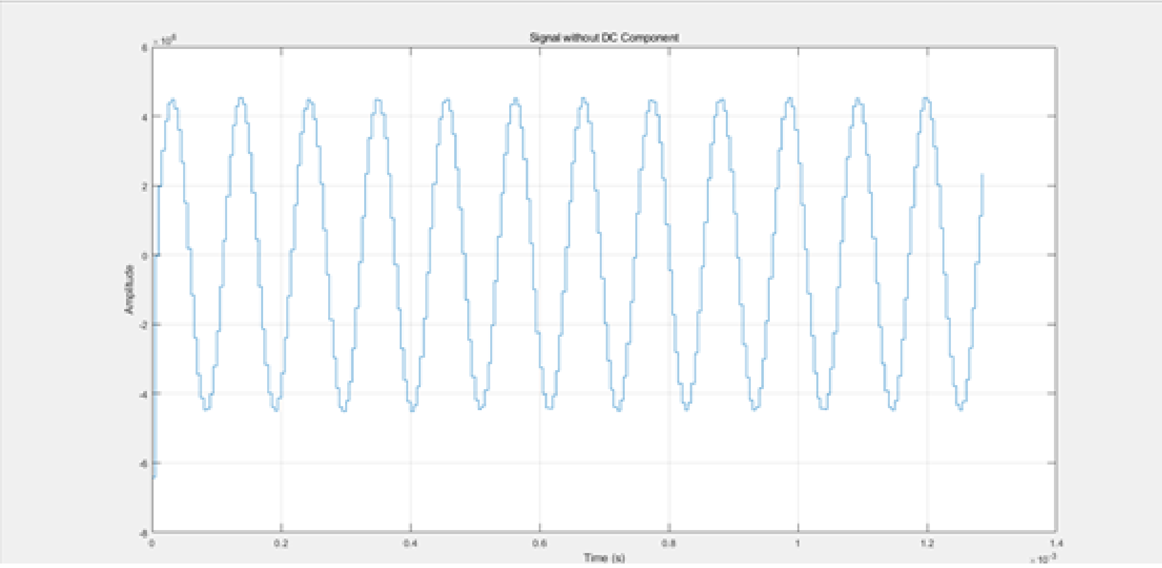

Figure 1.27: D/A result

FFT analysis of this waveform reveals the post-filtering spectrum:

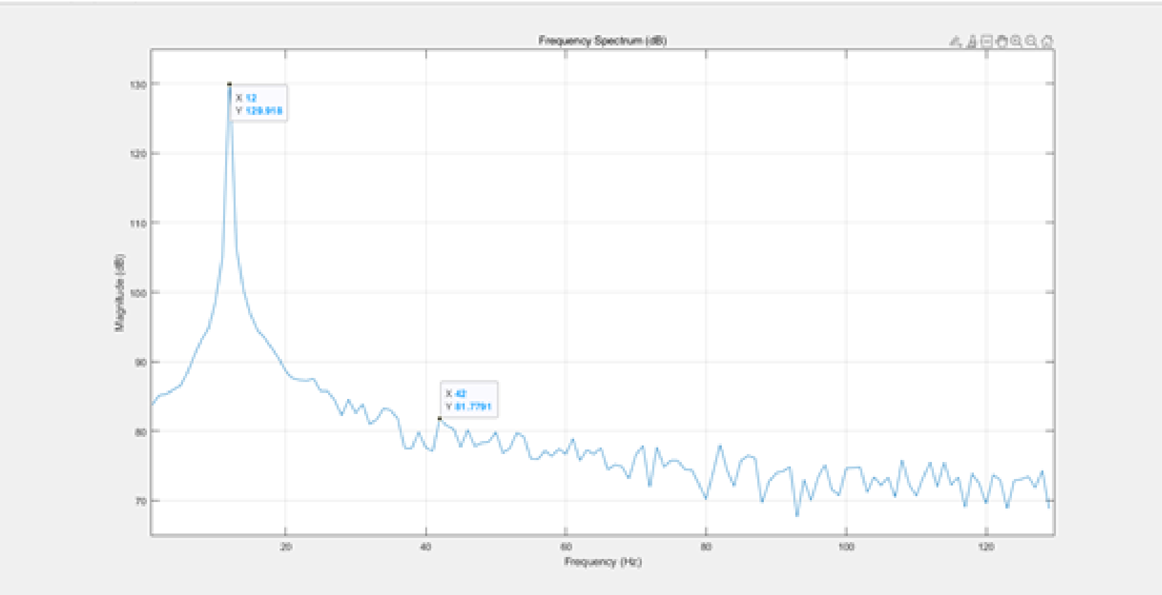

Figure 1.28: Output Spectrum

The resulting SNR is approximately 48 dB with the decimation filter at nominal.

1.2.5 How does temperature affect your SNR? Simulate your system at different temperature corners (range -40°C – 125°C). Is your system still functionable?

Fig.1.29 shows the variation of output spectrum across temperature. Clearly the converter is functional across the simulated temperature ranges. Table 1.3 shows the results measured with the simulated temperature corner.

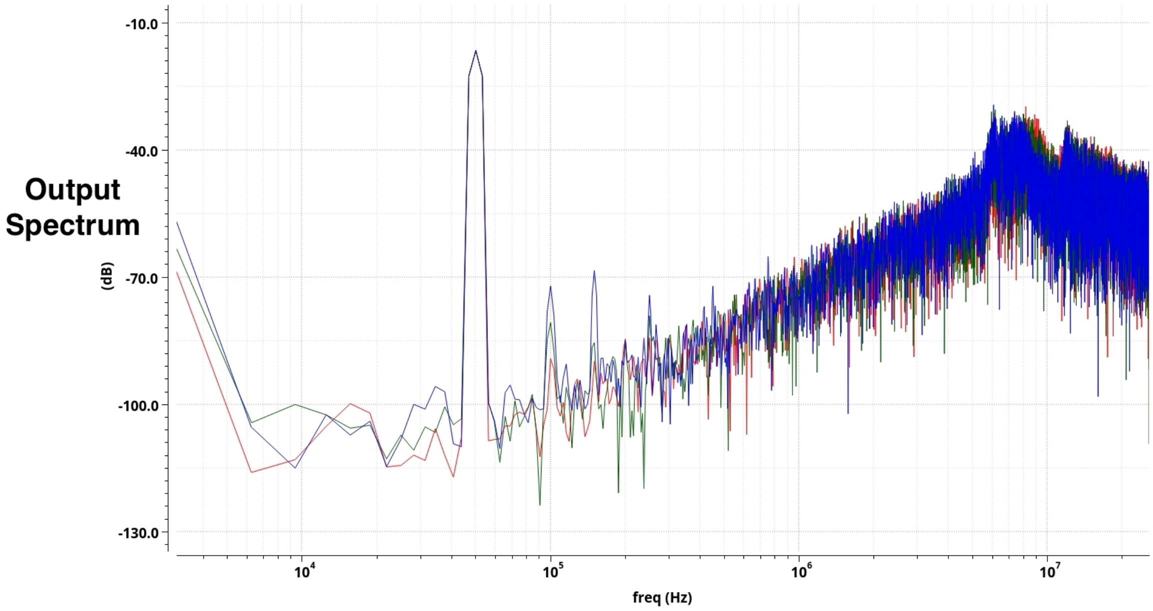

Figure 1.29: FFT Spectrum of the 2^{nd} order Δ-Σ Modulator Designed in Virtuoso for -40^0C and 125^0C

Temperature	SNR (dB)	ENOB	SFDR (dB)	Power (µW)
27^0C	63.57	10.27	64.14	230
125^0C	55.74	8.96	55.49	222.4
-40^0C	71.49	11.58	72.68	231.6

Table 1.3: Temperature variation Metrics

1.2.6 How does supply voltage's variation affects your converter's performance? Expect variation around ±10%.

Fig.1.31 shows the variation of output spectrum across supply. Clearly the converter is functional in the ±10%*V_{DD} range. Table 1.4 shows the results measured with the simulated Supply voltaga corner.

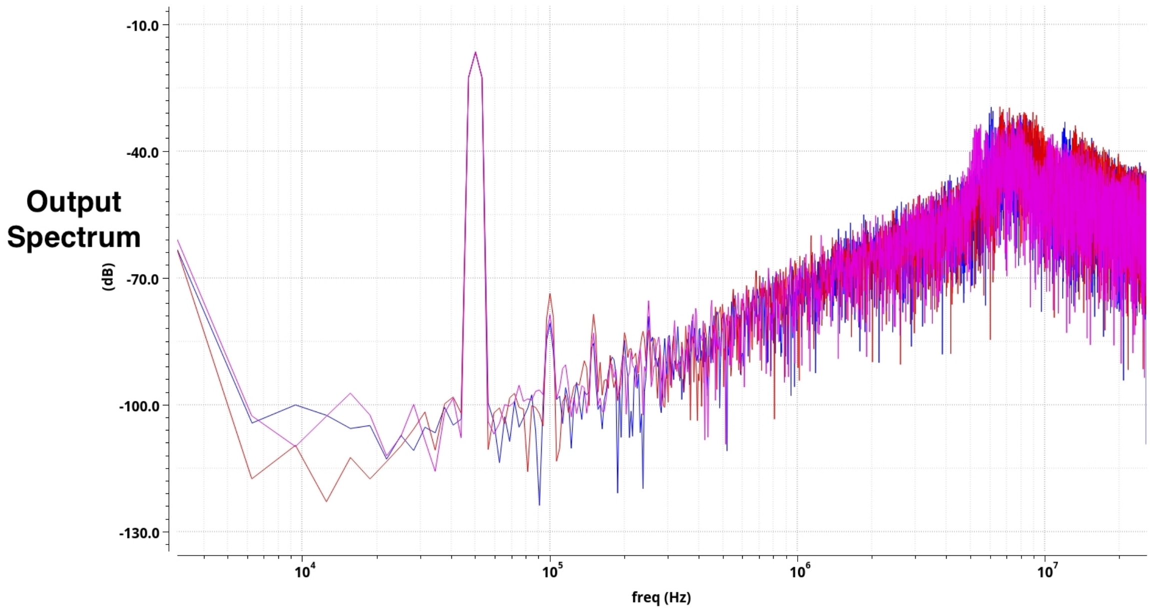

Figure 1.30: FFT Spectrum of the 2^{nd} order Δ-Σ Modulator Designed in Virtuoso for V_{DD} of ±605mV and ±495mV

Supply Voltage	SNR (dB)	ENOB	SFDR (dB)	Power (µW)
±550mV	63.57	10.27	64.14	230
±605mV	57.32	9.22	57.09	323.3
±495mV	61.61	9.94	62.07	159.7

Table 1.4: Voltage Variation Metrics

1.2.7 Investigate the SNR at different input frequencies.

Table 1.5 shows the results measured with variable input frequencies. Fig.?? shows the spectrum at an input frequency of 100KHz.

Input Frequency	SNR (dB)	ENOB	SFDR (dB)
50KHz	63.57	10.27	64.14
10KHz	69.3	11.22	71.2
100KHz	72.69	11.78	77.96

Table 1.5: Input Frequency Variation Metrics

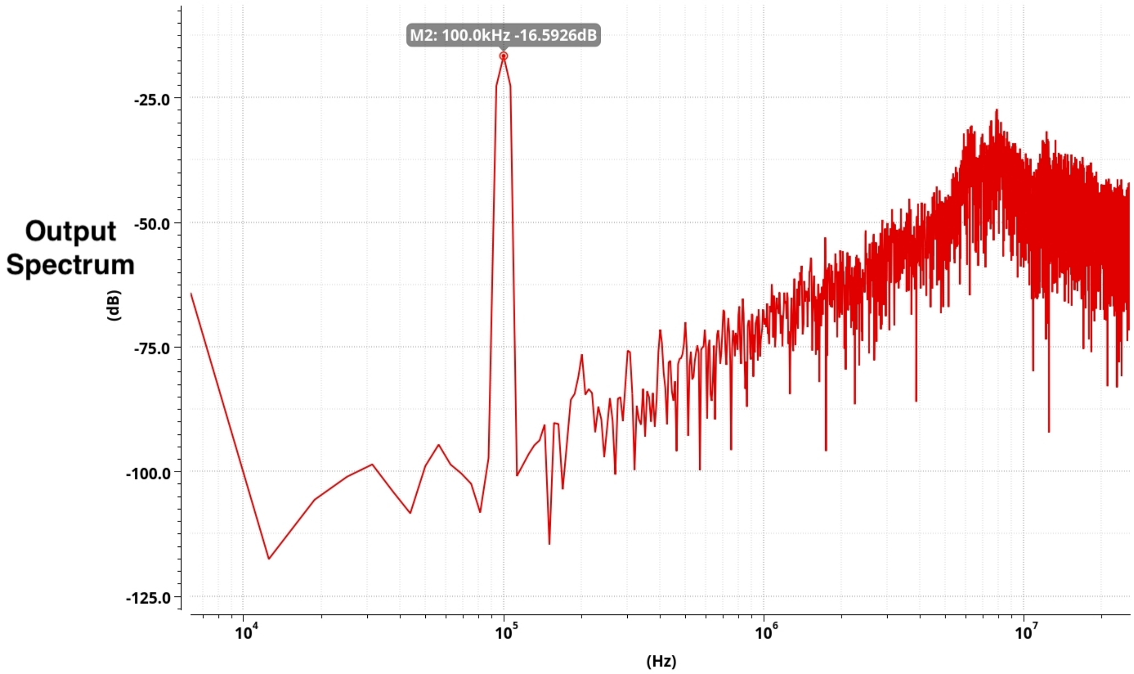

Figure 1.31: FFT Spectrum of the 2^{nd} order Δ-Σ Modulator Designed in Virtuoso for Input Frequency of 100KHz

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